Faculty of Engineering & Information Technology
University of Technology, Sydney

31338 Network Servers
Lab 1a: VMware Lab familiarisation

Aims

1. To explore the VMware software

2. To be able to start and shutdown your image running under VMware

Task 1: Navigate through the VMware Lab

Install VMware following the Lab set up tutorial and download *CentOS* and *Windows Server* system images. (i.e. ova files)

Import system images. (A USB drive or SSD with over 80GB of space is recommoned.)

On your lab workstation, choose "Applications -> System Tools -> VMplayer" to start the VMware player.

From VMware player/workstation, click Open a Virtual Machine ->select CentOS/WindowsServer.ova -> Provide the virtual machine's name and storage path, -> click import.

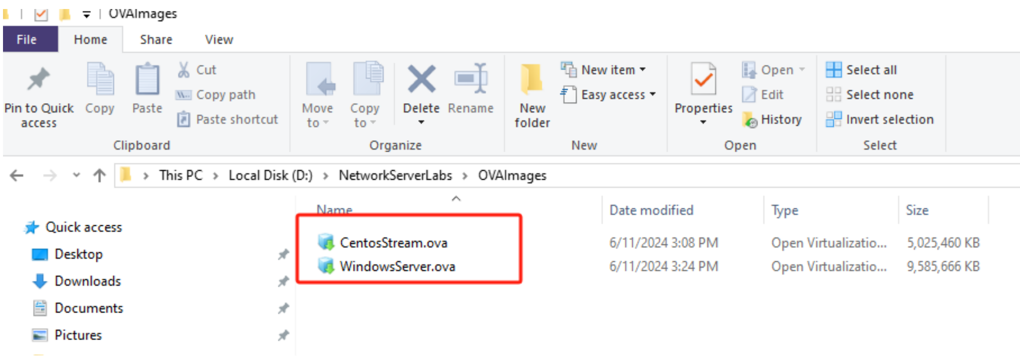
Note that the path could be:

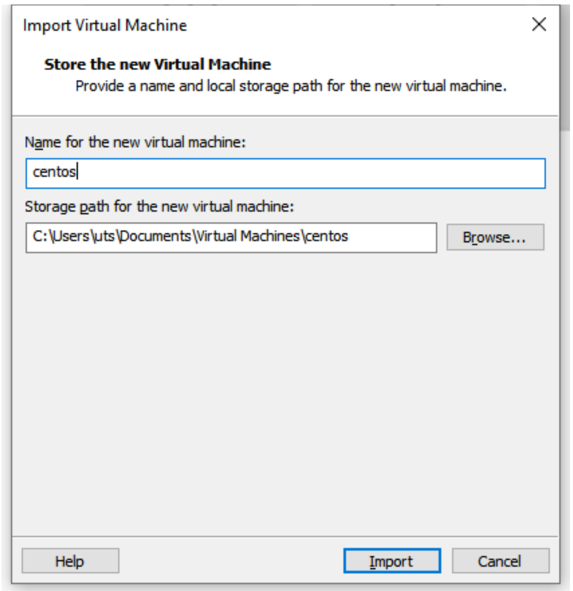
/class/NetServers/home/nnnnnnnn/centos64 (where nnnnnnn is your student number) [LAN LAB ONLY!]

or

/media/XYZ/centos64 (where XYZ is the label assigned to your USB hard disk)

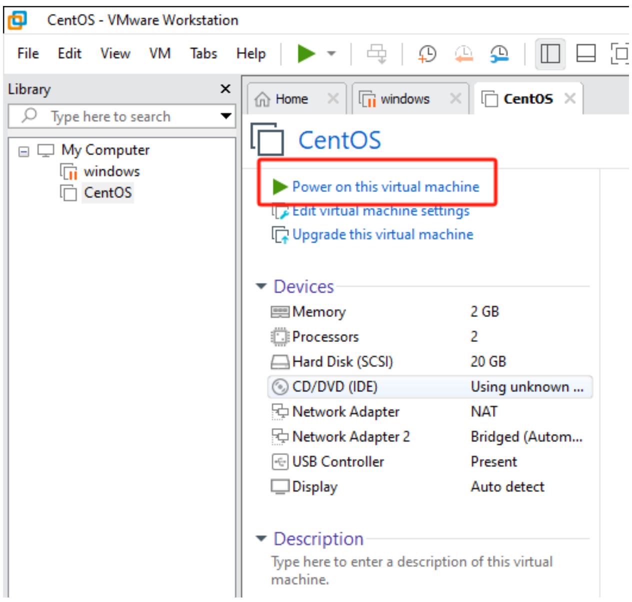
You should only ever need to do "Import" once.





Starting VMware systems:

Once you have imported the systems, click "Power on this virtual machine" to start.



If you move to another machine with your USB/SSD, From VMware Player, “File → Open “ and select the relevant directory of your USB hard disk -> Choose the .vmx file appropriate to your virtual machine (normally there is only 1)

You should now see the startup panel – the state, OS, Version, RAM will be displayed.

Select the “Edit VM settings” button.

Explore the devices listed here - what do you think they are and what should you (or can you) change?

As part of a system administrator’s role, you should document the configuration of your systems, so write the configuration down into your learning journal.

NOTE: In most organisations, recording hardware will be done via a formal process somewhere - either though an enterprise configuration manager, a formal database, spreadsheets etc. Many organisations still use paper…. And so will we!

To get mouse/keyboard access to your VM, click a mouse inside the window.

HINT: when in console mode, press the Ctrl-Alt key combination to return to your desktop

You should see the following occur.

1. BIOS startup – this flashes up quite quickly!

Optional task: If you want to see the setup screen, then As SOON as you see the logo, press F2 to enter the BIOS setup screen. You can examine the settings eg: time and date & boot up sequence.

2. When the GRUB logo appears, you can either let the system boot the default system (Linux) OR

Optional Task: press any key (eg: enter) QUICKLY to get the GRUB menu. This is where you can change the startup parameters of your operating system. What parameters can you see on the Linux configuration?

3. When the system starts booting, you should see a progress bar on the bottom of the screen – press ESC key to see the actually bootup sequence. For now, just observe the startup activities for the Linux bootup

<u>Task 2: Logging in as root and shutting down your Linux image</u>

Let's log in as the root user (superuser). In the username field, type: "root" and press Enter. When it asks you for a password, type:

student123! (The root password should never be this simple in real life!!)

You can go through the 'gnome-initial-setup' where it asks you for languages, keyboards, location services, and your online / social media accounts (DON'T CONNECT THEM). At each stage to move to the next step, press the blue "Next" or grey "Skip" button at the top right of the window.

Feel free to start exploring your system a little if you like. But please do not change the password.

You can get a terminal by selecting "Activities" and then choosing the picture that looks like a grey command window.

From this command prompt, you can check the network by typing ifconfig

What do you see? Can you determine if there is a default tcpip address?

Try recall your various Unix commands from Web Systems/Unix Systems Programming.

How about the ls command?

Logging on as root

It is an extremely rare practice to logon to root. We have provided a trivial password only because this is a "disposable" lab. In real life, the root password should be very hard to guess and crack because this has superman powers and you can easily destroy your system. Be careful – it is very easy to cripple or destroy an installation when working as root.

The best practice is to logon as yourself, and to use the sudo command to execute privileged commands. We will look at this later, so the 2nd best way to do this is to use the "superuser" command su

Type:

su

You will be prompted for the password. Enter it. Observe that the prompt changes to a # symbol to remind you that you have the superuser power

Reboot the system.

There are various commands to do this, but the simplest under these circumstances is reboot.

Why would a system admin not normally just reboot?

Observe the screen during the shutdown and rebooting process.

Make notes as appropriate in your Learning Journal. In general, you should answer questions (such as that above) in your Journal.

Then when you are ready, shut down your virtual machine with:

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shutdown –h now
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Again, document what happens.

NOTE: You normally do not have to do this - our Linux distribution allows us to shutdown or restart via the menus.

Choose the icon that looks like a power on/off button near the top right.

Why do you think it is important to learn to shutdown from the command line?

Answer this in your learning journal…